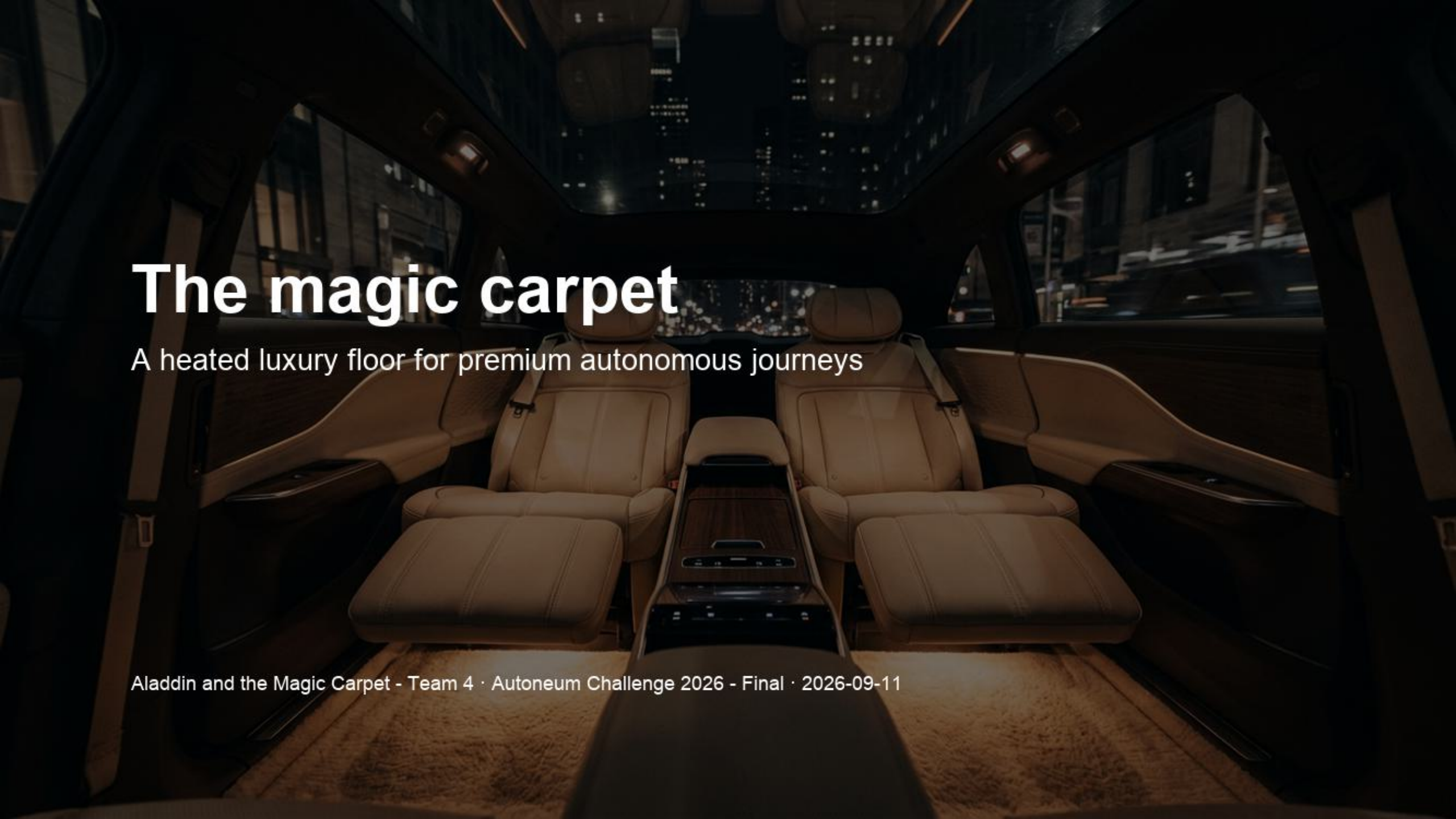
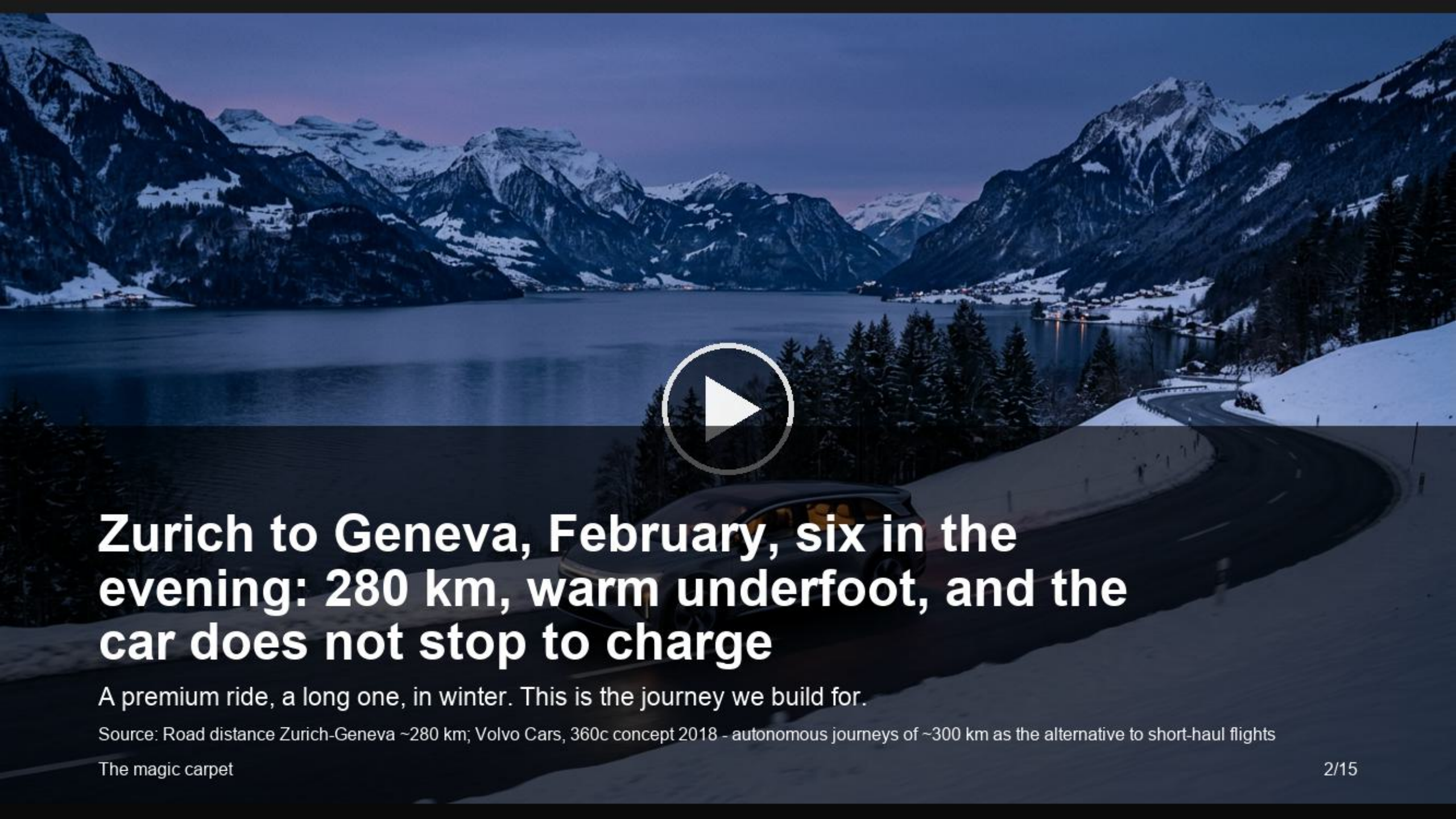


The image shows the interior of a vehicle, likely a luxury autonomous car. The view is from the passenger side looking towards the driver's seat. There are two rows of light-colored leather seats with horizontal stitching. The floor is covered in a light-colored, textured material, possibly a heated carpet. The ceiling is dark with some ambient lighting. The windows show a blurred city street at night.

The magic carpet

A heated luxury floor for premium autonomous journeys

Aladdin and the Magic Carpet - Team 4 · Autoneum Challenge 2026 - Final · 2026-09-11



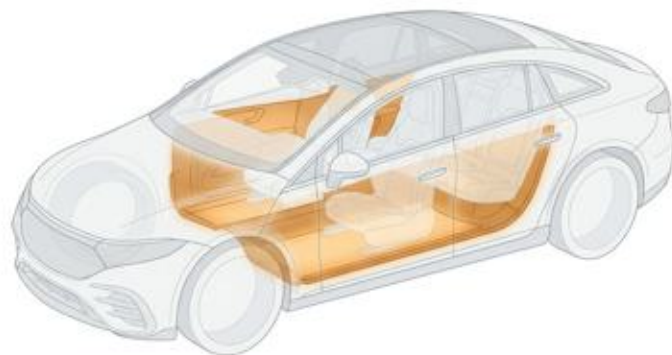
Zurich to Geneva, February, six in the evening: 280 km, warm underfoot, and the car does not stop to charge

A premium ride, a long one, in winter. This is the journey we build for.

Source: Road distance Zurich-Geneva ~280 km; Volvo Cars, 360c concept 2018 - autonomous journeys of ~300 km as the alternative to short-haul flights

The magic carpet

By 2035 the premium ride is a long one, and the fare pays for the cabin



USE CASE - THE TARGET

Premium ride services

Uber Black class, long journeys. The fare pays for the cabin, not the distance.

SEGMENT

Premium BEV, Level 4

Europe and Switzerland - Autoneum's largest market, 46 % of revenue.

AND A BONUS

The private owner

Runs the same floor at the weekend. More on him at the end.

Source: Goldman Sachs, Robotaxis to become a \$400 bn market in 2035 (Apr 2026); Autoneum FY2025 revenue release (Jan 2026)

Heating the air is the single largest winter drain on an electric car

RESISTIVE CABIN HEATING

3-5 kW

continuous, the largest cold-weather load

RANGE AT -7 C, HEATING ON

-41 %

AAA test across five EVs

SAME COLD, HEATING OFF

-12 %

the cabin is the difference

Source: AAA, Cold weather reduces electric vehicle range, Feb 2019 (20 F / -7 C, five vehicles); US DOE VTO program record, Sep 2024

THE UNFAIR FIGHT

On a 280-km journey that drain is a charging stop - and a premium fare waiting at a charger is a lost customer

A car with 400 km of nominal range keeps about 240 with the heater on. Zurich to Geneva is 280.

Source: AAA 2019 (-41 % with heating at -7 C); road distance Zurich-Geneva ~280 km

The magic carpet

Blown air is the wrong tool for a lounge

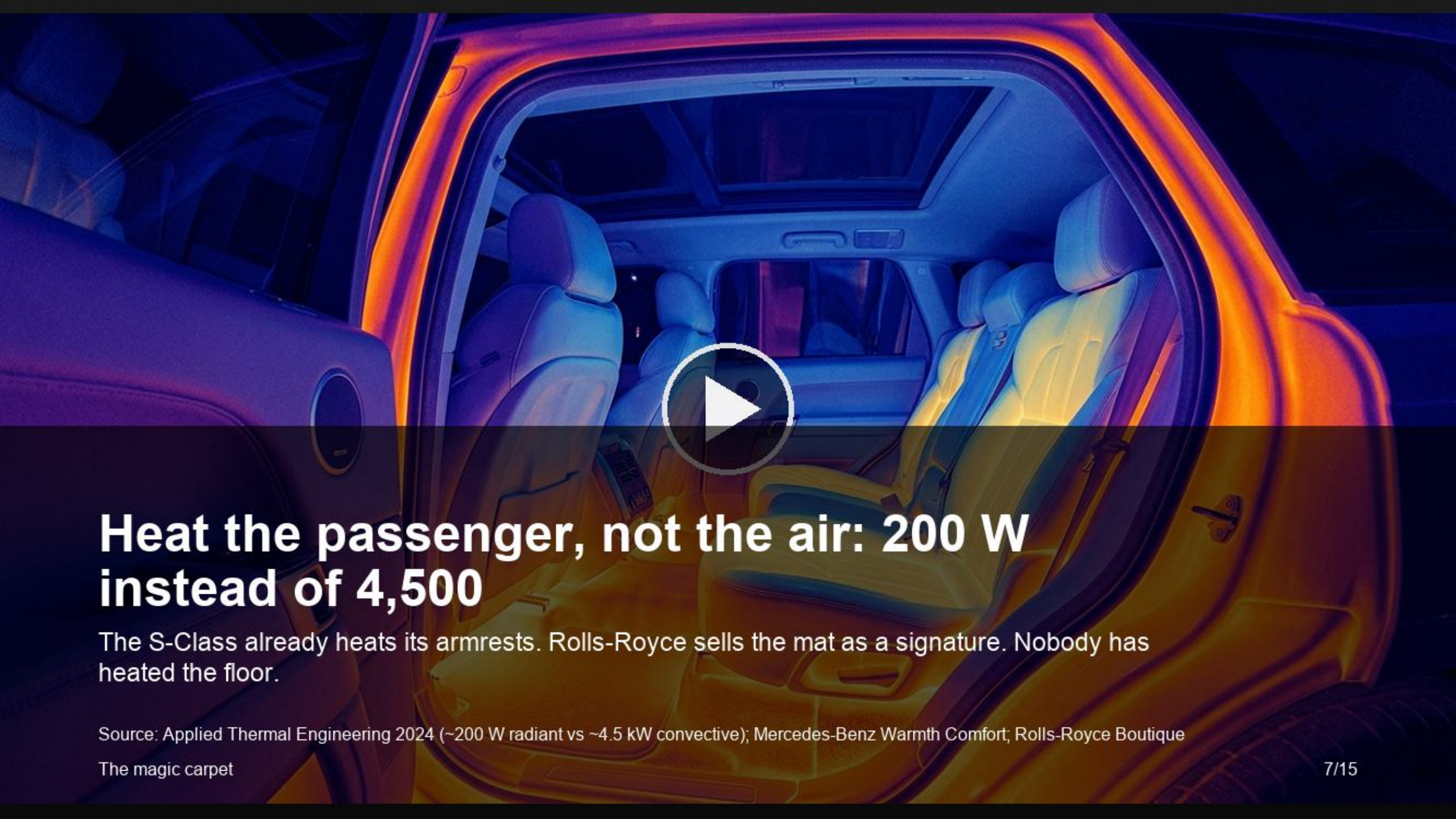
Climate control is built around a dashboard. The dashboard is gone.

- No fixed seats, so no vent points at anyone
- Dry, noisy, and gone at every door
- Two warm surfaces: neutral comfort 6 C lower



Source: Energy and Buildings 2026 - neutral comfort ~6 C below setpoint with two radiant sources

The magic carpet



Heat the passenger, not the air: 200 W instead of 4,500

The S-Class already heats its armrests. Rolls-Royce sells the mat as a signature. Nobody has heated the floor.

Source: Applied Thermal Engineering 2024 (~200 W radiant vs ~4.5 kW convective); Mercedes-Benz Warmth Comfort; Rolls-Royce Boutique

The magic carpet

THE HARDER PROBLEM

But a heated floor is either loud or cold

An acoustic absorber is a thermal insulator. Heater under it: the warmth goes into the body. Heater on top: it blocks the sound.

Autoneum already makes every layer of it

The layer	Autoneum has	Where it sits today
Heating layer	(Heated) floor mats	Trucks
Absorber	Hybrid-Acoustics PET	Interior floor
Coated yet acoustic	Alpha-Liner	Wheelhouse, outside
Absorb through a closed skin	RIMIC micro-perforation	Heatshields
Premium surface	Di-Light, Relive-1	Interior floor

Source: Autoneum Challenge deck, 7 Sep 2026, slides 6-7; Autoneum product brochure 2021

The build: a premium skin, a heater that lets sound through, an absorber, a dock

Perforated premium skin

Lambswool look, wipeable, sound passes through

Printed heating tracks

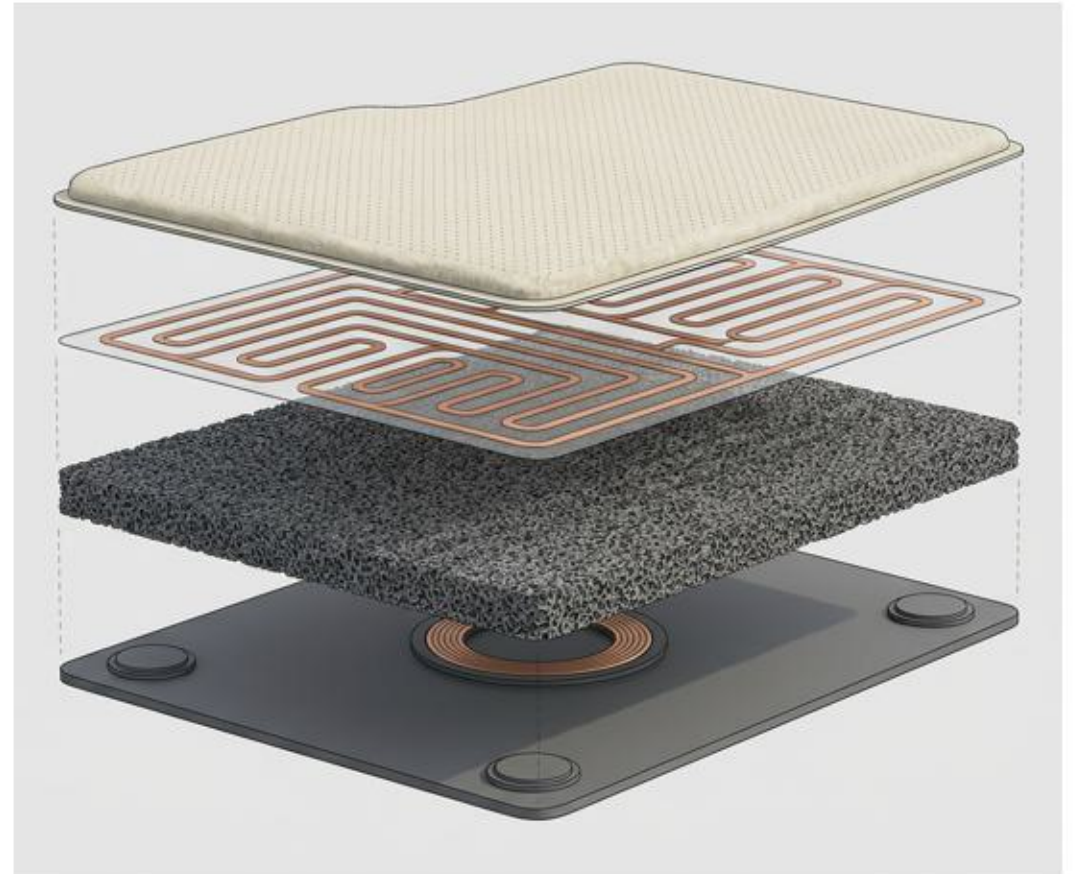
On top of the absorber, acoustically transparent

Absorber

Hybrid-Acoustics PET - and it insulates downwards

Magnetic dock

Contactless power, no visible cable



The premium market is being built right now

Not a concept. Fleets, prices, and buyers on record.

UBER X LUCID X NURO

20,000

luxury-SUV robotaxis from 2026

WAYMO'S BASE CAR

USD 70,000

a Jaguar, since 2018

RIDERS PAY

+13-40 %

Waymo over Uber, 2025-26

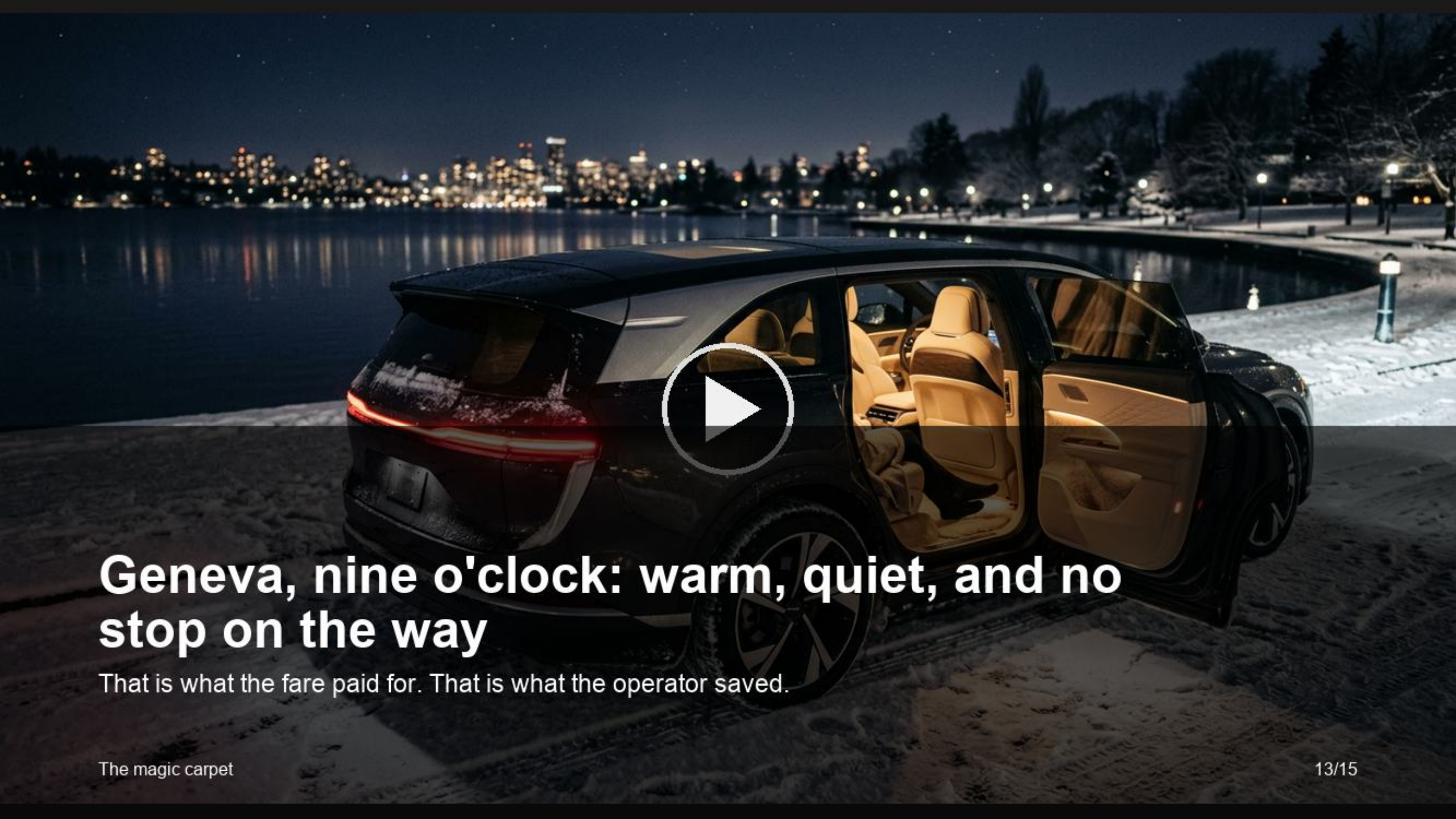
Source: Lucid / Uber / Nuro, Jul 2025; Waymo Blog 2018; Obi via TechCrunch, Jan 2026; McKinsey CFM Consumer Survey 2022

A luxury surface has to stay luxury - so it lifts out in seconds

Serviced off the car. Never scrubbed inside it.

- Lambswool for winter, linen-look for summer
- Magnetic dock, no tools, thirty seconds
- The same dock in a private car - later





Geneva, nine o'clock: warm, quiet, and no stop on the way

That is what the fare paid for. That is what the operator saved.

ONE MORE THING

**The private owner runs the same floor
at the weekend - and a standard one
when the car earns**

Same dock, thirty seconds. He comes free with the platform.

What we ask the board for

- A pilot: one premium fleet, fifty heated floors, one winter
- Lab time: thermography and an impedance tube
- Treat the dock as the platform and the heated floor as its first product

Sources: Autoneum challenge deck 2026, product brochure 2021, FY2025 release; AAA 2019, US DOE 2024; Applied Thermal Engineering 2024, Energy and Buildings 2026; Goldman Sachs 2026; Lucid-Uber-Nuro 2025; Obi / TechCrunch 2026; Mercedes-Benz, Rolls-Royce, Volvo. Full references in the presenter script.